IXYS

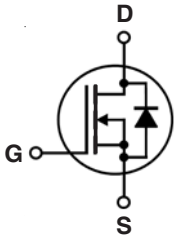
Depletion Mode MOSFET

IXTY01N100D
IXTU01N100D
IXTP01N100D

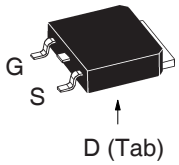
V_{DSX} = 1000V

$R_{DS(on)} \leq$ 80Ω

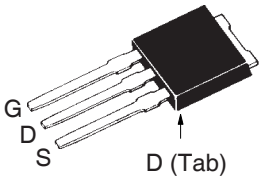
N-Channel



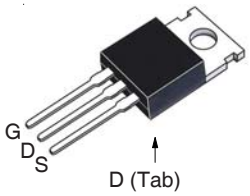
TO-252 (IXTY)



TO-251 (IXTU)



TO-220 (IXTP)



G = Gate D = Drain
S = Source Tab = Drain

Symbol	Test Conditions	Maximum Ratings	
V_{DSX}	T_J = 25°C to 150°C	1000	V
V_{DGX}	T_J = 25°C to 150°C	1000	V
V_{GSX}	Continuous	±20	V
V_{GSM}	Transient	±30	V
I_{DM}	T_C = 25°C, Pulse Width Limited by T_J	400	mA
P_D	T_C = 25°C	25	W
	T_A = 25°C	1.1	W
T_J		- 55 ... +150	°C
T_{JM}		150	°C
T_{stg}		- 55 ... +150	°C
T_L	Maximum Lead Temperature for Soldering	300	°C
T_{SOLD}	1.6 mm (0.062in.) from Case for 10s	260	°C
M_d	Mounting Torque (TO-220)	1.13 / 10	Nm/lb.in.
Weight	TO-252	0.35	g
	TO-251	0.40	g
	TO-220	3.00	g

Symbol	Test Conditions (T_J = 25°C, Unless Otherwise Specified)	Characteristic Values Min.	Typ.	Max.	
BV_{DSX}	V_{GS} = -10V, I_D = 25μA	1000			V
$V_{GS(off)}$	V_{DS} = 25V, I_D = 25μA	- 2.0		- 4.5	V
I_{GSX}	V_{GS} = ±20V, V_{DS} = 0V			±100	nA
$I_{DSX(off)}$	V_{DS} = V_{DSX}, V_{GS} = -10V			10	μA
	T_J = 125°C			250	μA
$R_{DS(on)}$	V_{GS} = 0V, I_D = 50mA, Note 1		50	80	Ω
$I_{D(on)}$	V_{GS} = 0V, V_{DS} = 25V, Note 1		400		mA

Features

- Normally ON Mode
- International Standard Packages
- Low $R_{DS(on)}$ HDMOS™ Process
- Rugged Polysilicon Gate Cell Structure
- Fast Switching Speed

Advantages

- Easy to Mount
- Space Savings
- High Power Density

Applications

- Level Shifting
- Triggers
- Solid State Relays
- Current Regulators

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Symbol	Test Conditions (T_J = 25°C, Unless Otherwise Specified)	Characteristic Values Min.	Typ.	Max.	
g_{fs}	V_{DS} = 100V, I_D = 100mA, Note 1	100	200		mS
C_{iss}	V_{GS} = -10V, V_{DS} = 25V, f = 1MHz		100		pF
C_{oss}			12		pF
C_{rss}			2		pF
$t_{d(on)}$	**Resistive Switching Times**		7		ns
t_r	V_{GS} = ±5V, V_{DS} = 50V, I_D = 50mA		10		ns
$t_{d(off)}$	R_G = 30Ω (External)		34		ns
t_f			64		ns
$Q_{g(on)}$	V_{GS} = ±5V, V_{DS} = 500V, I_D = 50mA		5.8		nC
Q_{gs}			3.6		nC
Q_{gd}			0.4		nC
R_{thJC}				5.0	°C/W
R_{thCS}	TO-220		0.50		°C/W

Source-Drain Diode

Symbol	Test Conditions (T_J = 25°C, Unless Otherwise Specified)	Characteristic Values Min.	Typ.	Max.	
V_{SD}	I_F = 100mA, V_{GS} = -10V, Note 1			1.5	V
t_{rr}	I_F = 750mA, -di/dt = 100A/µs V_R = 25V, V_{GS} = -10V			1.5	µs

Note 1. Pulse test, t ≤ 300µs, duty cycle, d ≤ 2%.

IXYS Reserves the Right to Change Limits, Test Conditions, and Dimensions.

IXYS MOSFETs and IGBTs are covered by one or more of the following U.S. patents: 4,835,592 4,860,072 4,881,106 4,931,844 5,017,508 5,034,796 5,049,961 5,063,307 5,187,117 5,237,481 5,381,025 5,486,715 6,162,665 6,259,123 B1 6,306,728 B1 6,404,065 B1 6,534,343 6,583,505 6,683,344 6,710,405 B2 6,710,463 6,727,585 6,759,692 6,771,478 B2 7,005,734 B2 7,063,975 B2 7,071,537 7,157,338B2

Fig. 1. Output Characteristics @ T_J = 25°C

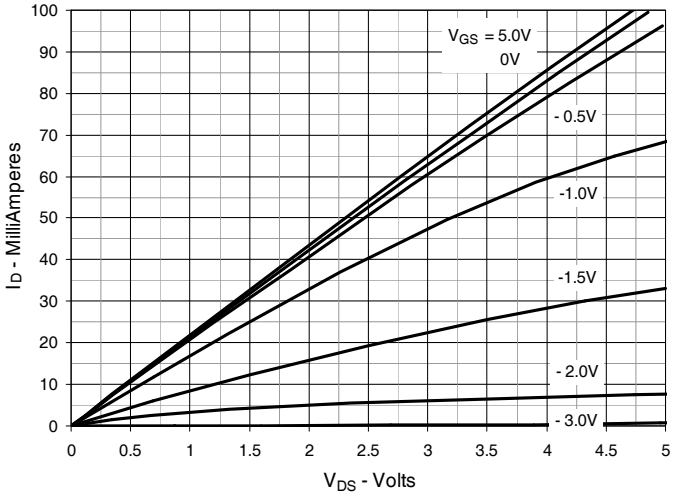


Fig. 2. Output Characteristics @ T_J = 125°C

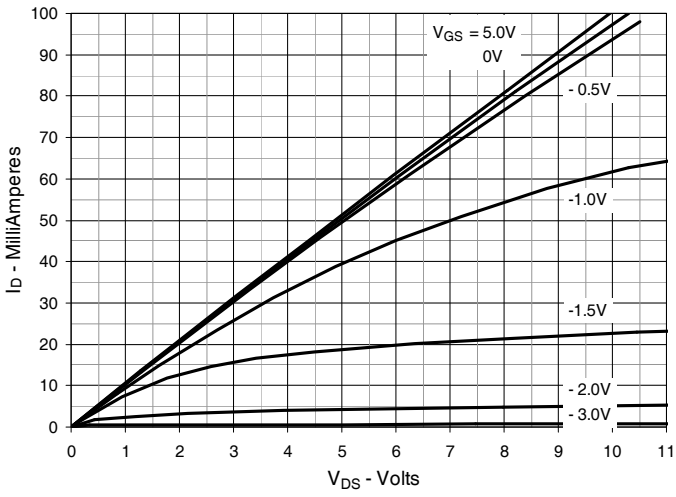


Fig. 3. Drain Current @ T_J = 25°C

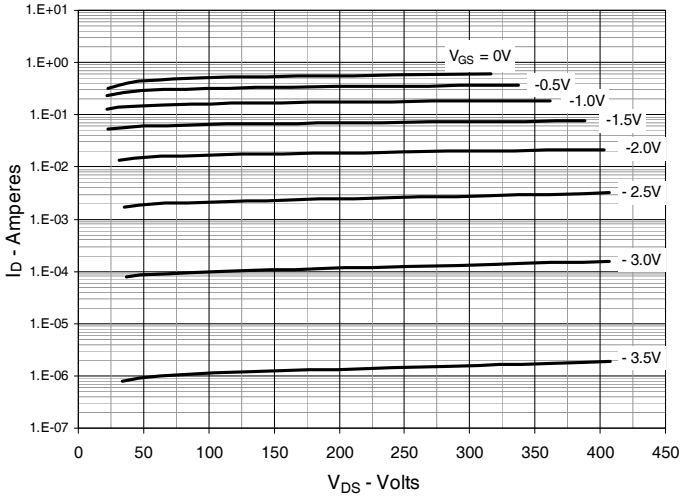


Fig. 4. Drain Current @ T_J = 100°C

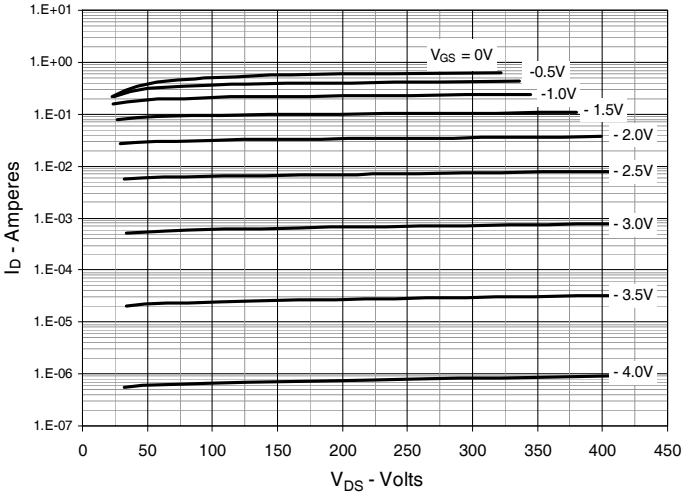


Fig. 5. Dynamic Resistance vs. Gate Voltage

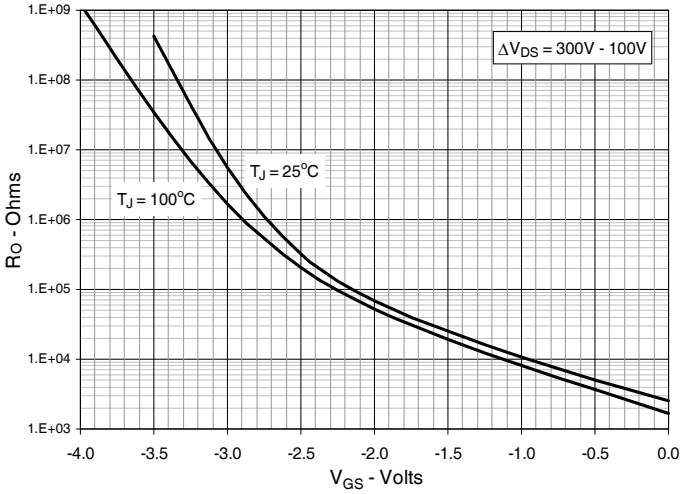
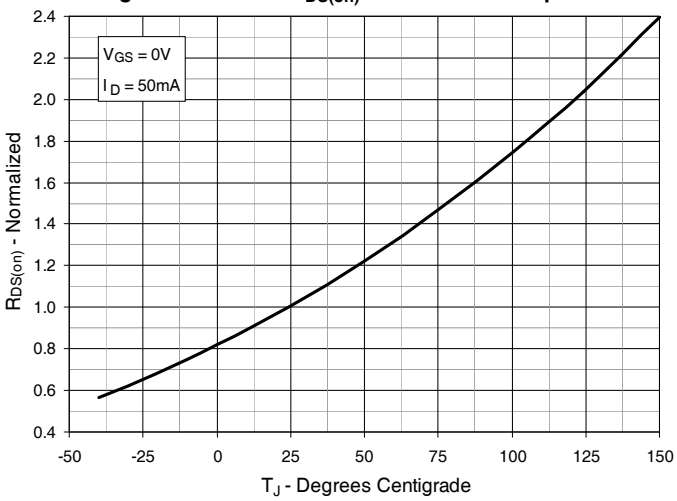


Fig. 6. Normalized $R_{DS(on)}$ vs. Junction Temperature



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Fig. 7. $R_{DS(on)}$ Normalized to I_D = 50mA Value vs. Drain Current

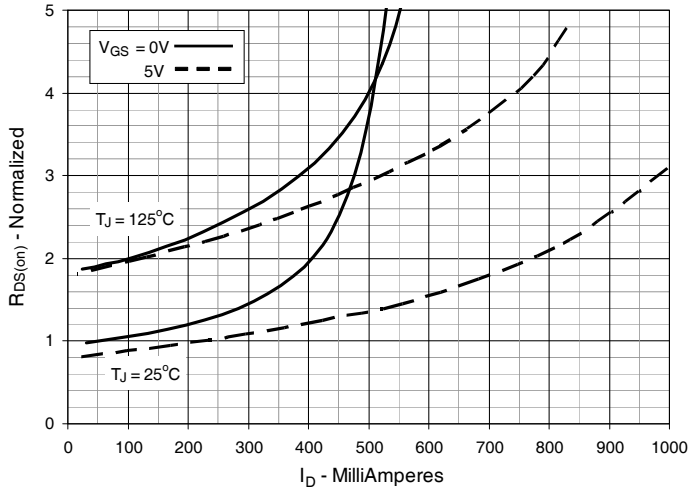


Fig. 8. Input Admittance

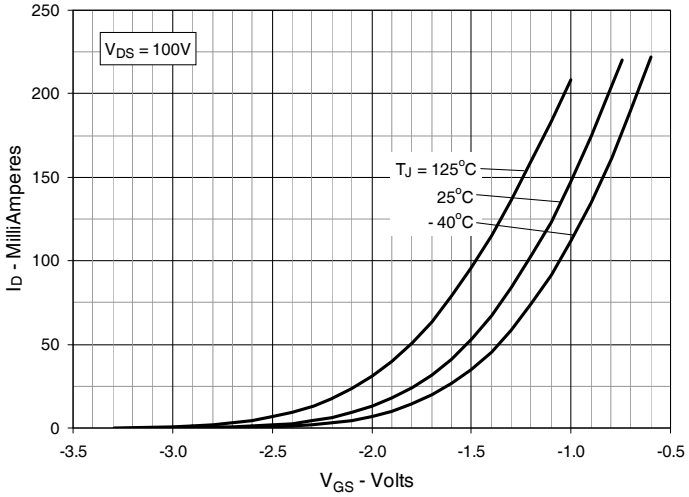


Fig. 9. Transconductance

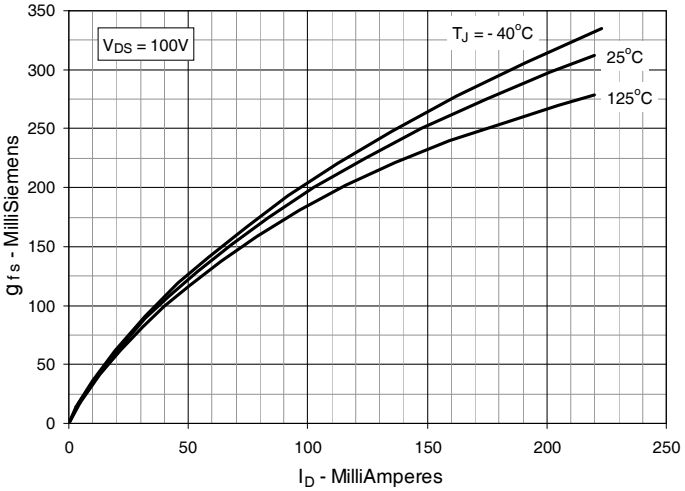


Fig. 10. Forward Voltage Drop of Intrintsic Diode

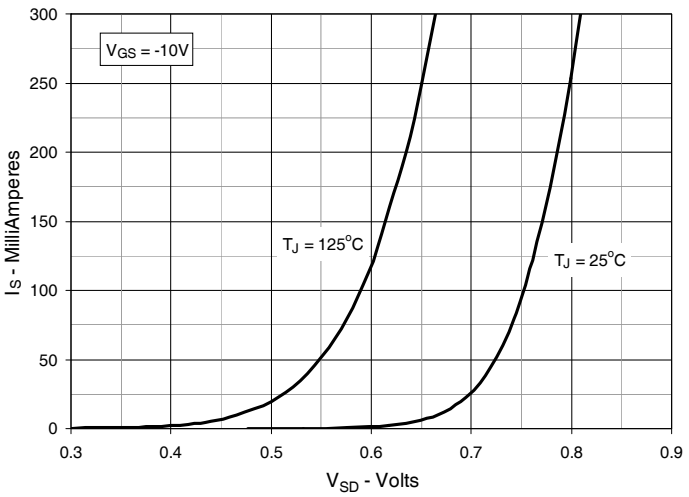


Fig. 11. Capacitance

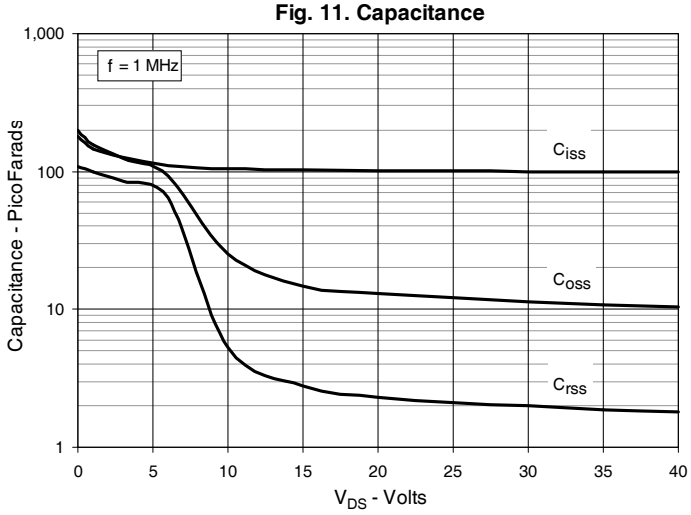
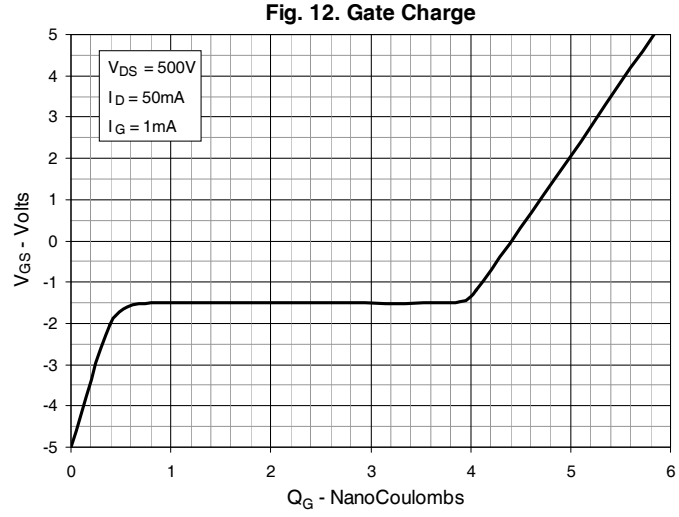


Fig. 12. Gate Charge



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Fig. 13. Forward-Bias Safe Operating Area @ T_C = 25°C

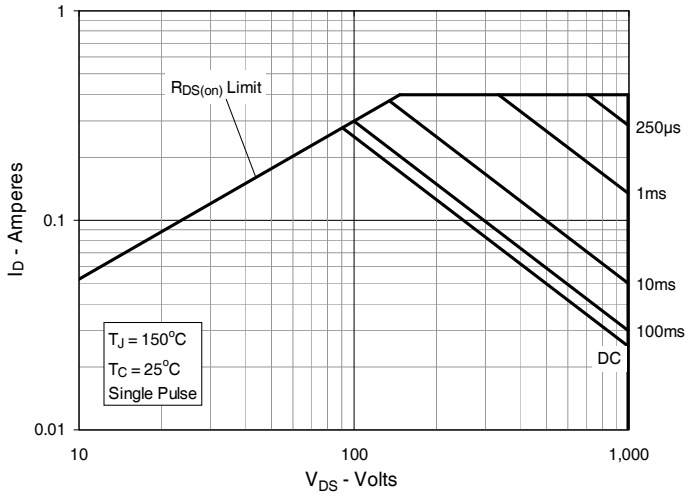


Fig. 14. Forward-Bias Safe Operating Area @ T_C = 75°C

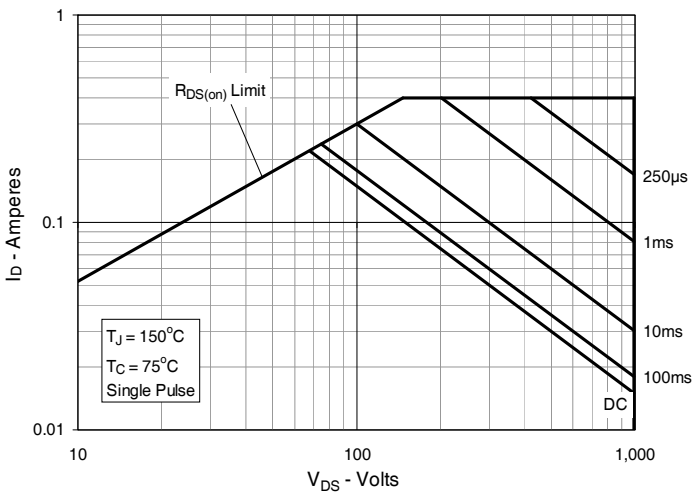
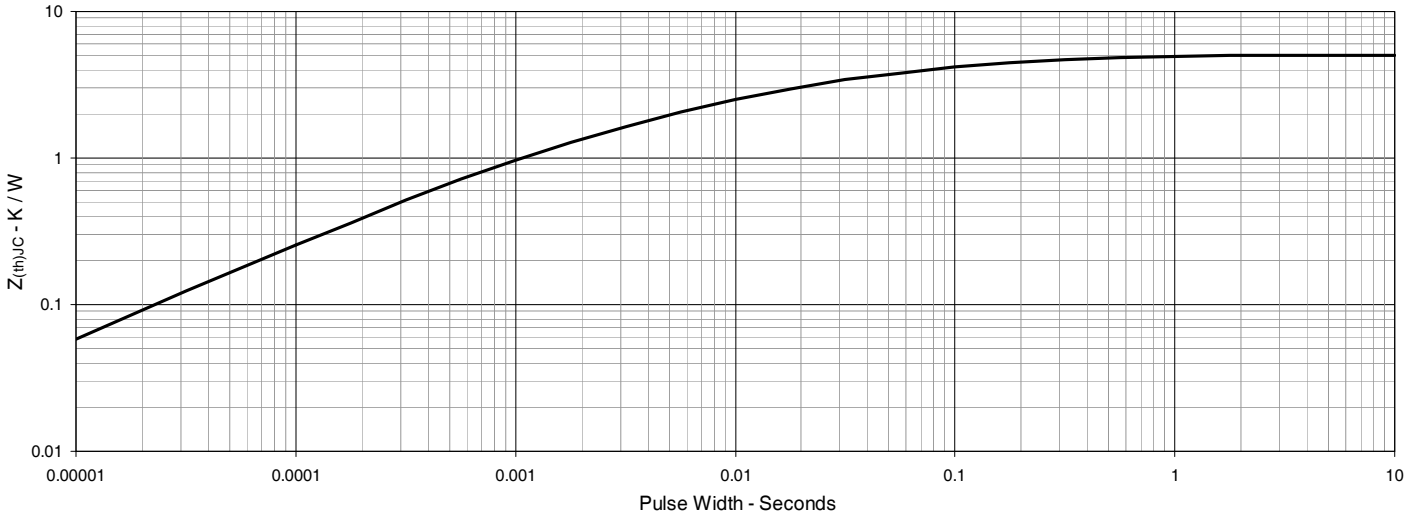
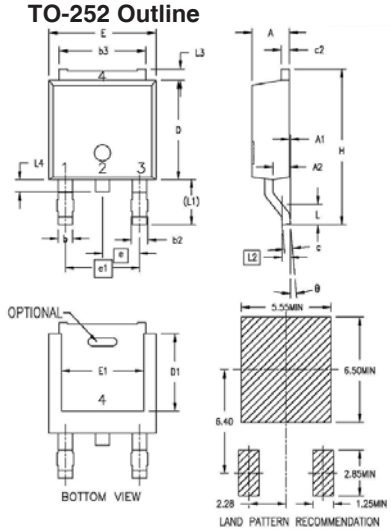


Fig. 15. Maximum Transient Thermal Impedance



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TO-252 Outline

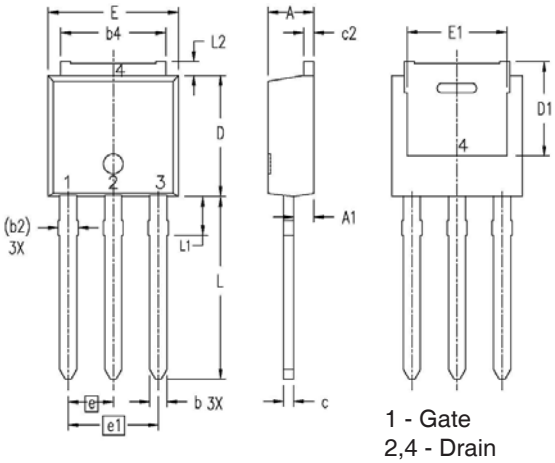


1 - Gate
2,4 - Drain
3 - Source

SYM	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.086	.094	2.19	2.38
A1	0	.005	0	0.12
A2	.038	.046	0.97	1.17
b	.025	.035	0.64	0.89
b2	.030	.045	0.76	1.14
b3	.200	.215	5.08	5.46
c	.018	.024	0.46	0.61
c2	.018	.023	0.46	0.58
D	.235	.245	5.97	6.22
D1	.180	.205	4.57	5.21
E	.250	.265	6.35	6.73
E1	.170	.205	4.32	5.21
e	.090 BSC		2.28 BSC	
e1	.180 BSC		4.57 BSC	
H	.370	.410	9.40	10.42
L	.055	.070	1.40	1.78
L1	.100	.115	2.54	2.92
L2	.020 BSC		0.50 BSC	
L3	.025	.040	0.64	1.02
L4	.025	.040	0.64	1.02
θ	0°	10°	0°	10°

NOTE: 1. This drawing comply JEDEC TO-252AA value except L3 dimension.
2. All metal surface are tin plated except trimmed area.

TO-251 Outline

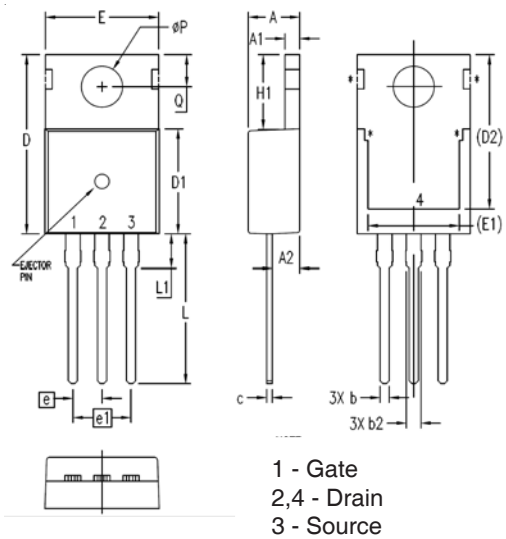


1 - Gate
2,4 - Drain
3 - Source

SYM	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.087	.094	2.20	2.40
A1	.032	.048	0.82	1.22
b	.026	.034	0.66	0.86
(b2)	.030	.038	0.76	0.96
b4	.198	.222	5.04	5.64
c	.018	.024	0.45	0.60
c2	.016	.024	0.40	0.60
D	.232	.248	5.90	6.30
(D1)	.179	.195	4.55	4.95
E	.252	.268	6.40	6.80
(E1)	.191	.207	4.85	5.25
e	.090 BSC		2.28 BSC	
e1	.180 BSC		4.57 BSC	
L	.358	.374	9.10	9.50
L1	.063	.079	1.60	2.00
L2	.020	.035	0.50	0.90

NOTE: 1. ALL METAL AREA ARE MATTE PURE TIN PLATED EXCEPT TRIMMED AREA.
2. THESE DIMENSIONS DO NOT INCLUDE PROTRUSIONS OF THE MOLD.
3. THE ( ) MARK IS THE REFERENCE ONLY.

TO-220 Outline



1 - Gate
2,4 - Drain
3 - Source

SYM	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.169	.185	4.30	4.70
A1	.047	.055	1.20	1.40
A2	.079	.106	2.00	2.70
b	.024	.039	0.60	1.00
b2	.045	.057	1.15	1.45
c	.014	.026	0.35	0.65
D	.587	.626	14.90	15.90
D1	.335	.370	8.50	9.40
(D2)	.500	.531	12.70	13.50
E	.382	.406	9.70	10.30
(E1)	.283	.323	7.20	8.20
e	.100 BSC		2.54 BSC	
e1	.200 BSC		5.08 BSC	
H1	.244	.268	6.20	6.80
L	.492	.547	12.50	13.90
L1	.110	.154	2.80	3.90
ØP	.134	.150	3.40	3.80
Q	.106	.126	2.70	3.20

NOTE:
1. All metal surface are matte pure tin plated except trimmed area.



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